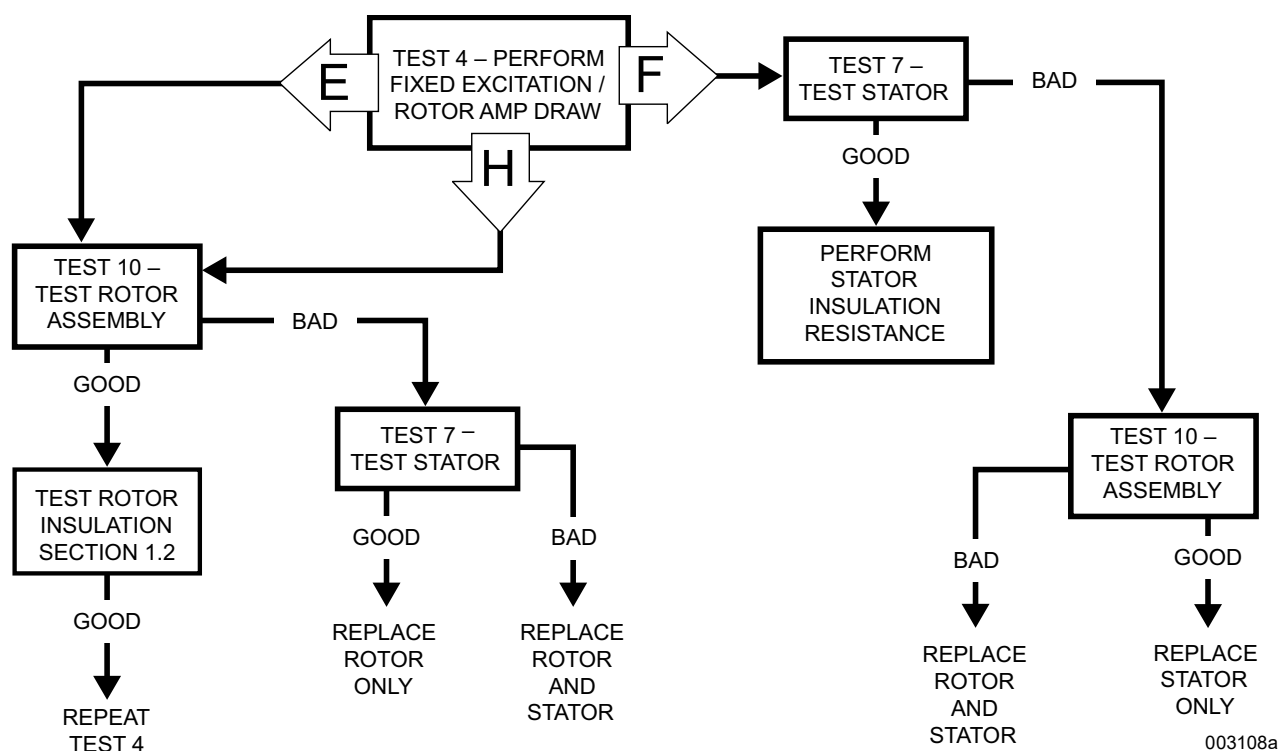
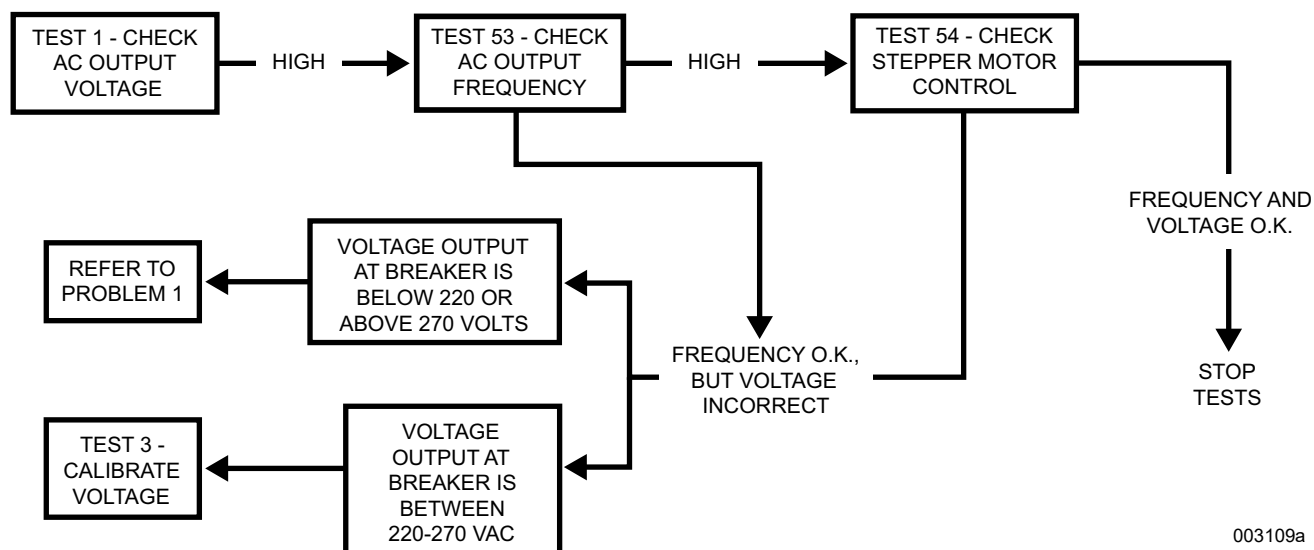


Problem 1 – Generator Shuts Down for Under Voltage (Continued)

NOTE: Evolution 1.0 (non-Synergy) Firmware (v1.12 and above) and Evolution 2.0 (all firmware), perform the Preliminary Output Voltage Test.

**Problem 2 – Generator Produces High Voltage**

IMPORTANT NOTE: Clear any faults in the controller before proceeding with any running diagnostic steps!

Results

1. If the DMM indicated approximately 240-244 VAC, the output voltage is good.
2. If the DMM indicated any other readings the voltage is bad. Refer back to the flow chart.

NOTE: “Residual” voltage may be defined as the voltage produced by rotor residual magnetism alone. The amount of voltage induced into the stator AC power windings by residual voltage alone will be approximately 2 to 16 VAC, depending on the characteristics of the specific generator. If a unit is supplying residual voltage only, either excitation current is not reaching the rotor or the rotor windings are open and the excitation current cannot pass. On current units with air-cooled engines, “field boost” current flow is available to the rotor after the engine reaches 2200 rpm.

Test 3 – Calibrate Voltage

General Theory

When voltage output is too high or too low, it is possible to adjust voltage output of the generator. To access this menu a password will be required to be entered into controller.

NOTE: Replacement controllers must be checked but typically do not require calibration unless output is not within the specifications. (Refer to the unit data decal and to the Specifications section in the front of this manual.)

Procedure

1. Set DMM to measure AC voltage.
2. Open the Main Line Circuit Breaker (MLCB) on the generator.
3. On the controller, press the ESC key until the main menu is present. (See [Evolution Menu System Navigation](#))
4. From the main menu enter the dealer password and proceed to DEALER > DEALER EDIT > CALIBRATE VOLTAGE then press ENTER.
5. After pressing enter, a value will appear on the screen.
6. Using appropriate back probes, measure output voltage at the AVR connector (Wires 11S and 44S).

NOTE: The default setting from the factory for calibration is 1024. The Evolution controller is adjustable from 750-1280.

7. Set controller to MANUAL and allow engine to start.
8. While the unit is running, use the UP or DOWN arrows to adjust the calibration setting. A higher value will create a lower voltage. A lower value will create a higher voltage.
9. Once a desired output voltage has been achieved, press ENTER to save the new setting.

NOTE: The Calibration Setting will reset to being a password protected option after the controller is left idle.

Verification

While the unit is running, verify that the output voltage at the breaker is consistent within 5 volts of the value displayed in the controller TEST menu (See [Evolution Menu System Navigation](#)).

Results

1. If during the verification process the output voltage at the breaker and the display match and the calibration setting was not adjusted outside of the window, stop testing.
2. If a correct voltage output was not achieved using the window specified, perform the Field Boost Test (FBT) and then refer to “Problem 1 – Test 4 Fixed Excitation / Rotor Amp Draw Test.”

Field Boost Circuit Tests

Test FB1 – Wire 4 Test

General Theory

This test is to verify that Wire 4 is receiving field flash voltage from wire 4A during startup of the generator.

NOTE: See [Field Boost](#) in Section 2.1 for analysis of field boost parameters.

Procedure (using J3 Breakout Harness)

1. Set DMM to measure DC voltage.
2. Disconnect the generator harness J3 connector from the controller.
3. Connect the female end of the J3 breakout harness to the unit harness.

IMPORTANT NOTE: DO NOT CONNECT the male end of the J3 breakout harness to the controller.

4. Insert meter test leads into the DC excitation test points of the breakout harness (4 and 0).
5. Clear all faults and place generator in MANUAL mode and start engine.
6. Observe the meter for DC Field Flash Voltage.

Procedure (without using J3 Breakout Harness)

1. Set DMM to measure DC voltage.
2. Locate Wire 4 on back of control panel. Disconnect the wire harness connector from the control panel.
3. Insert the Red meter lead adapter into the back of the wire harness on Wire 4.
4. Place Black lead on a good common ground or negative post on the engine battery.
5. Clear all faults and place generator in the manual mode and start engine.
6. Measure and record field flash voltage.

NOTE: See [Field Boost](#) in Section 2.1 for analysis of field boost parameters.